DevOps & CI/CD Implementation

Objective

- Ensure reliable deployment of NLP framework
- Enable continuous integration of linguistic updates

Pipeline in This Project

- **Code Commit:** NLP modules (morphology, parser, API) via **GitHub**
- **Build: Integration of rule-based + ML components**
- **Test:**
  - **Morphological accuracy validation**
  - **API response testing (< 2 sec target)**
- **Package: Docker container** with NLP environment
- **Deploy: Cloud deployment** for research access

Project Insight

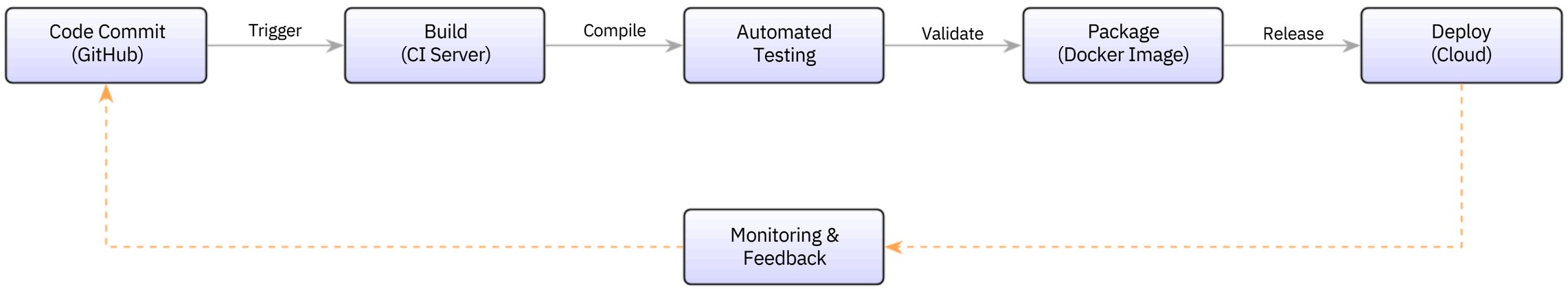
- Supports frequent updates of grammar rules (Panini sutras)
- Ensures consistent execution across environments
- Enables rapid bug fixes and feature releases

Management Impact

- **Reduces** deployment risk
- Improves delivery speed
- Enhances system reliability for stakeholders

DevOps & CI/CD Implementation

CI/CD Pipeline



Project Results & Outcomes

Project Performance

- **Total Cost: ₹33–41 Lakhs**
- **Project Duration: 218 Days (~10.5 Months)**
- **Payback Period: 3–4 Years**

Financial Evaluation

- **Net Present Value (NPV): Positive (~₹12.6 Lakhs)**
- **Indicates financial feasibility and long-term viability**

Operational Outcomes

- Achieved target accuracy (**~92%**) for analysis
- **Successful development of:**
  - **Morphological analyzer**
  - **Semantic parser**
  - **API-based system**

Management Outcome

- Delivered within planned scope, cost, and schedule
- **Effective use of:**
  - **Agile (Scrum)**
  - **Risk management**
  - **CI/CD practices**

Overall Impact

- Scalable platform for academic and research use
- Supports digital preservation and linguistic analysis

Conclusion

Project Summary

- Successfully planned and managed an AI-driven Sanskrit NLP framework
- Integrated traditional linguistic rules with modern technologies

Key Achievements

- Effective control of time, cost, and scope
- **Implementation of:**
  - **Structured planning (WBS, CPM)**
  - **Monitoring (EVA, Risk Management)**
  - **Agile & DevOps practices**

Managerial Insight

- Importance of **stakeholder collaboration (linguists + engineers)**
- Need for **adaptive planning in complex domains**
- Value of **continuous monitoring and iterative delivery**